

Lab 2

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Deadline: 2022-10-02 23:59:59, Beijing time

How to submit?

Email: YJ40202@163.com

Subject: [exercise](#)

Attachment: Your java source files.

Note: Don't use package.

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 Send

Recipient : YJ40202 <yj40202@163.com>  email address

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Exercise02_14.java

2.14 (Health application: computing BMI) Body Mass Index (BMI) is a measure of health on weight. It can be calculated by taking your weight in kilograms and dividing, by the square of your height in meters. Write a program that prompts the user to enter a weight in pounds and height in inches and displays the BMI. Note one pound is 0.45359237 kilograms and one inch is 0.0254 meters.

Here is a sample run:

```
Enter weight in pounds: 95.5 [Enter]  
Enter height in inches: 50 [Enter]  
BMI is 26.857257942215885
```

Exercise02_15.java

2.15 (Geometry: distance of two points) Write a program that prompts the user to enter two points (x_1, y_1) and (x_2, y_2) and displays their distance. The formula for computing the distance is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Note you can use `Math.pow(a, 0.5)` to compute $\sqrt{2}$.

Here is the sample run:

```
Enter x1 and y1: 1.5 -3.4 [Enter]
Enter x2 and y2: 4 5 [Enter]
The distance of the two points is 8.764131445842194
```

Exercise03_01.java

3.1 (Algebra: solve quadratic equations) The two roots of a quadratic equation $ax^2 + bx + c = 0$ can be obtained using the following formula:

$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$b^2 - 4ac$ is called the discriminant of the quadratic equation. If it is positive, the equation has two real roots. If it is zero, the equation has one root. If it is negative, the equation has no real roots.

Write a program that prompts the user to enter values for a, b, and c and displays the result based on the discriminant. If the discriminant is positive, display two roots. If the discriminant is 0, display one root. Otherwise, display "The equation has no real roots."

Exercise03_01.java

Here are some sample runs:

```
Enter a, b, c: 1 3 1 [Enter]
The equation has two roots -0.3819660112501051 and -2.618033988749895
```

```
Enter a, b, c: 1 2 1 [Enter]
The equation has one root -1.0
```

```
Enter a, b, c: 1 2 3 [Enter]
The equation has no real roots
```

Exercise03_05.java

3.05 (Find future dates) Write a program that prompts the user to enter an integer for today's day of the week (Sunday is 0, Monday is 1, . . . , and Saturday is 6). Also prompt the user to enter the number of days after today for a future day and display the future day of the week. Here is a sample run:

```
Enter today's day: 1 [Enter]
Enter the number of days elapsed since today: 3 [Enter]
Today is Monday and the future day is Thursday
```

Exercise03_12.java

3.12 (Palindrome integer) Write a program that prompts the user to enter a three-digit integer and determines whether it is a palindrome integer. An integer is palindrome if it reads the same from right to left and from left to right. A negative integer is treated the same as a positive integer. Here are sample runs of this program:

```
Enter a three-digit integer: 121 [Enter]
121 is a palindrome
```

```
Enter a three-digit integer: 123 [Enter]
123 is not a palindrome
```

Exercise03_18.java

3.18 (Cost of shipping) A shipping company uses the following function to calculate the cost (in dollars) of shipping based on the weight of the package (in pounds).

$$c(w) = \begin{cases} 3.5, & \text{if } w \leq 1 \\ 5.5, & \text{if } 1 < w \leq 3 \\ 8.5, & \text{if } 3 < w \leq 10 \\ 10.5, & \text{if } 10 < w \leq 20 \end{cases}$$

Write a program that prompts the user to enter the weight of the package and display the shipping cost. If the weight is greater than 20, display a message "the package cannot be shipped."

Exercise03_18.java

Here is the sample run:

```
Enter package weight: 1 [Enter]  
The shipping cost is $3.5
```

```
Enter package weight: 20 [Enter]  
The shipping cost is $10.5
```

Exercise03_33.java

3.33 (Financial: compare costs) Suppose you shop for rice in two different packages. You would like to write a program to compare the cost. The program prompts the user to enter the weight and price of each package and displays the one with the better price. Here is a sample run:

```
Enter weight and price for package 1: 50 24.59 [Enter]
Enter weight and price for package 2: 25 11.99 [Enter]
Package 2 has a better price.
```